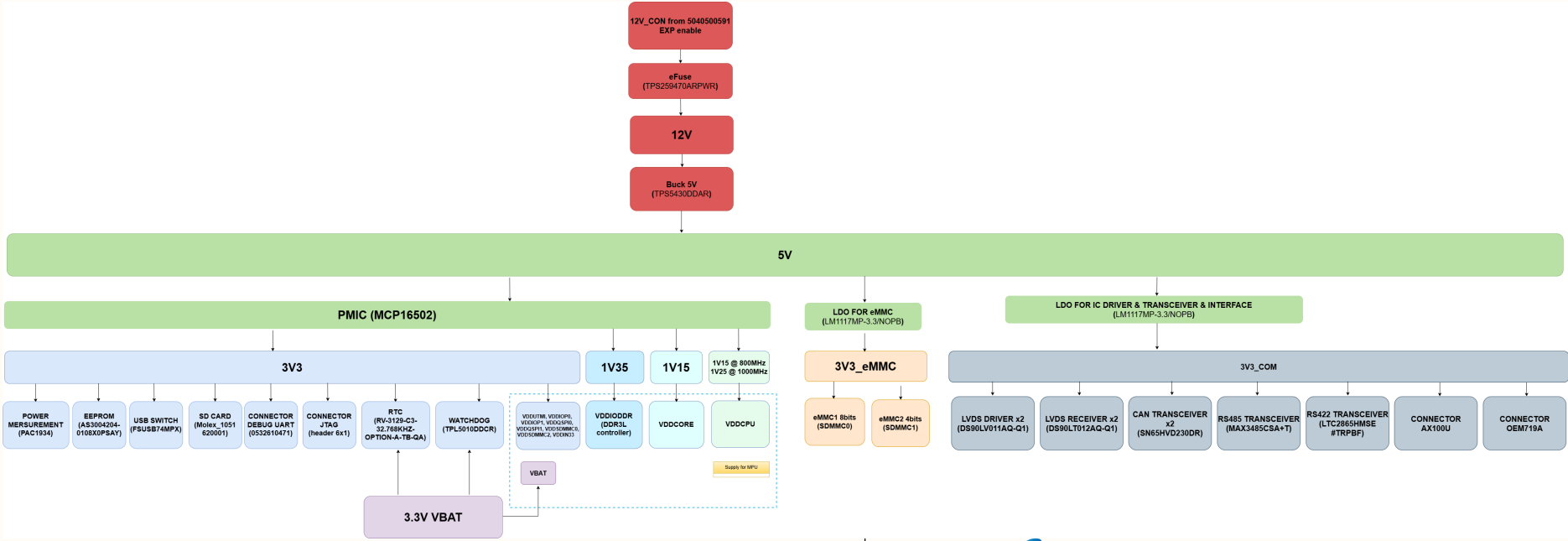
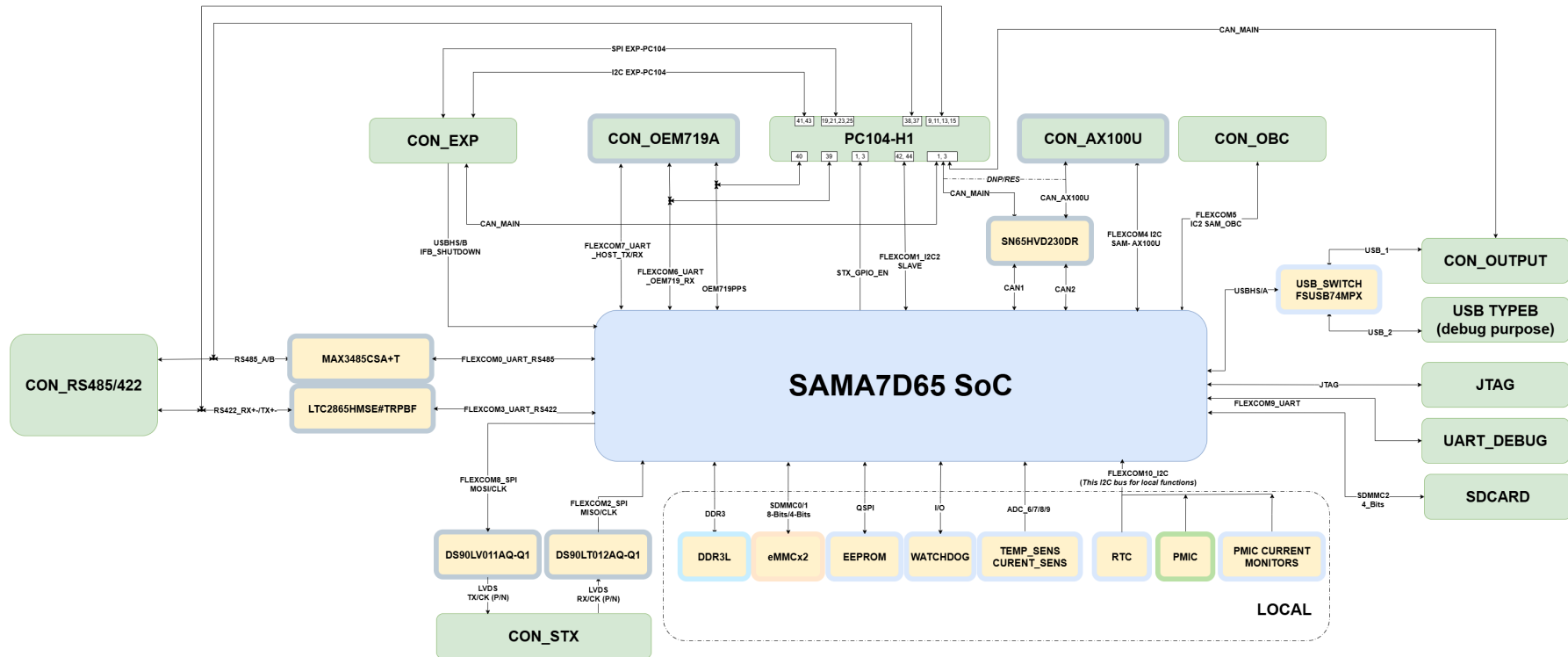




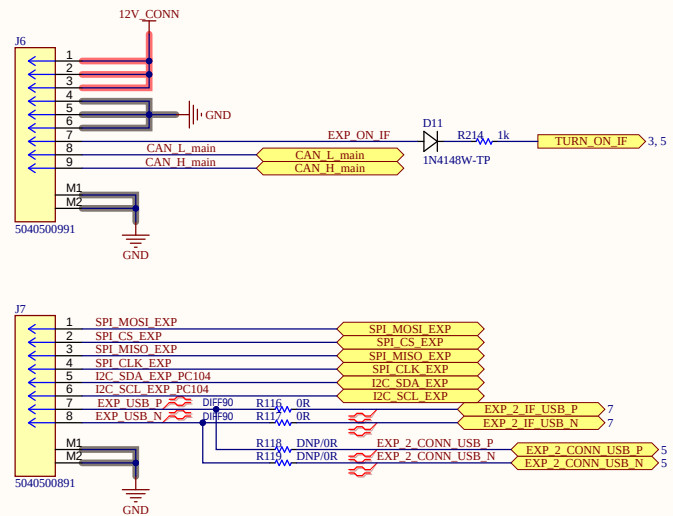
Project: BEE-1012		Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: PowerTree.SchDoc		Sheet 1 of 23
Checked by: BAO BUI	Document No: 1		Size: A3
Approved by: VU PHAM	Revision: *		Date: 1/27/2026

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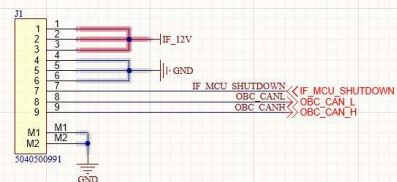


Project: BEE-1012		Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: BlockDiagram.SchDoc	Sheet 2 of 23	
Checked by: BAO BUI	Document No: 2	Size: A3	
Approved by: VU PHAM	Revision: *	Date: 1/27/2026	

Connect to EXP



form IF (OBC Board)



Project: **PAY_IF.PrjPcb**

Page Contents: **CONNECTOR_EXP.SchDoc**

Revision:

Design: *

Date: **2/3/2026**

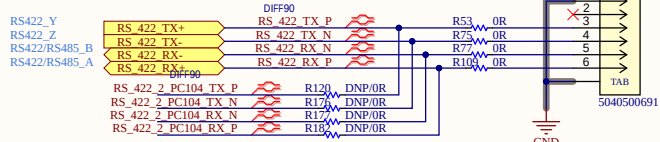
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Document Number:

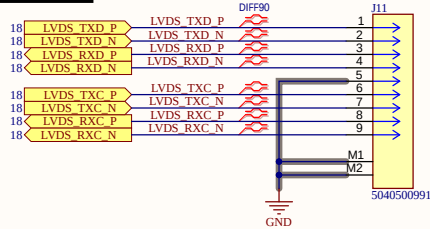
Sheet: **4** of **23**



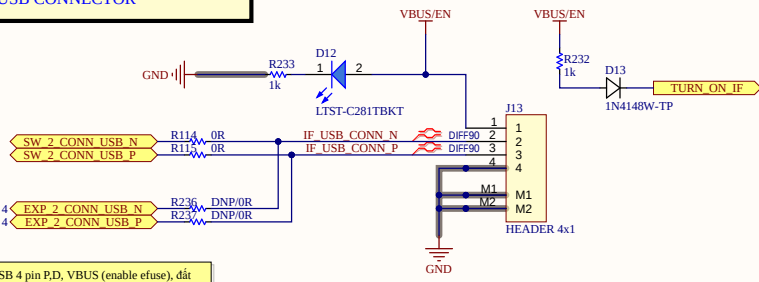
CONNECT RS485 RS422



CONNECT TO S_BAND TX

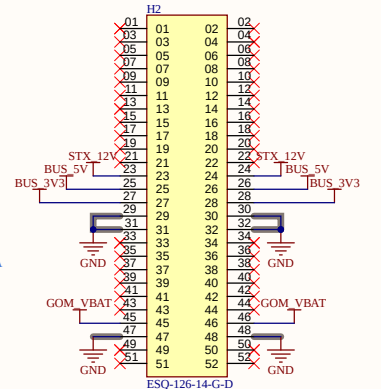
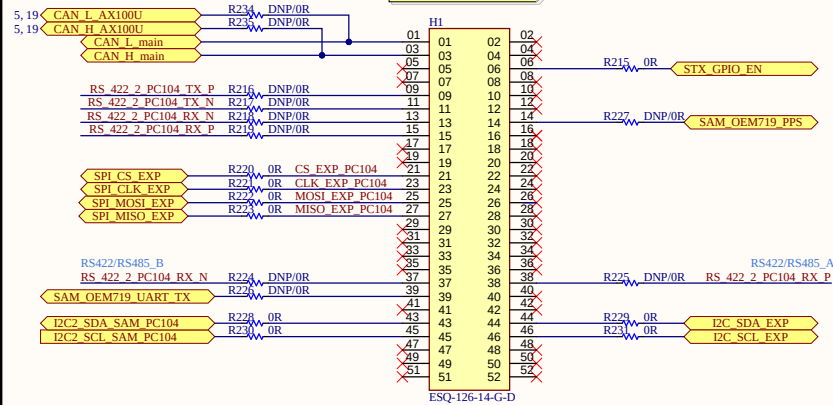


USB CONNECTOR

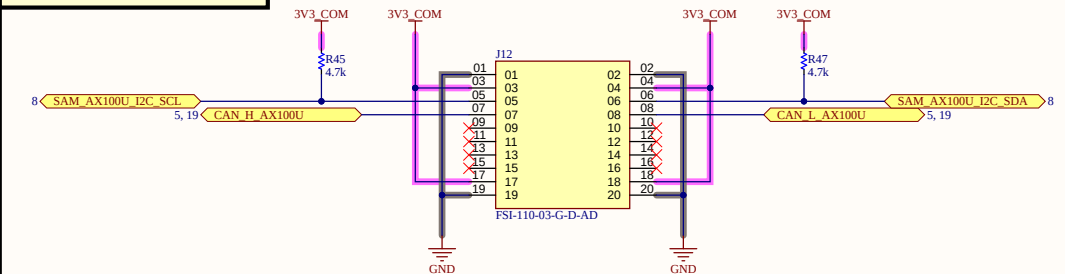


USB 4 pin P,D, VBUS (enable efuse), đất

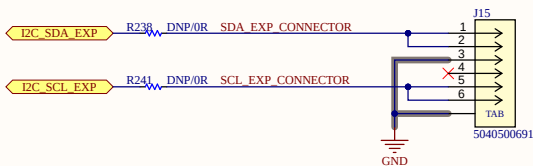
PC104



CONNECT TO AX100U



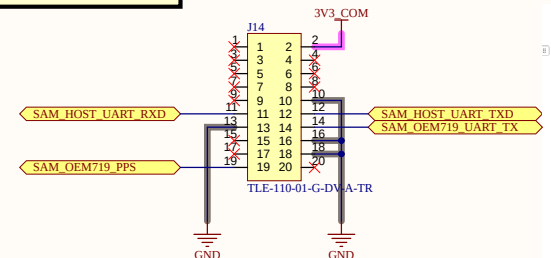
I2C CONNECTOR



SPI CONNECTOR



CONNECT TO OEM719A GNSS



Appendix C – OEM719A interface
Host connector for mating with OEM719A GNSS receiver

Source: TLE-110-01-G-DV-A-TR

Pin Description

Pin	Description
1	NC
2	3.3V
3	NC
4	NC
5	NC
6	NC
7	NC
8	NC
9	NC
10	NC
11	HOST_UART_RXD
12	HOST_UART_TXD
13	GND
14	DEMP19_UART2_TXD
15	NC
16	GND
17	NC
18	GND
19	OEM719_PPS
20	NC

Project: **PAY_IF.PrjPcb**

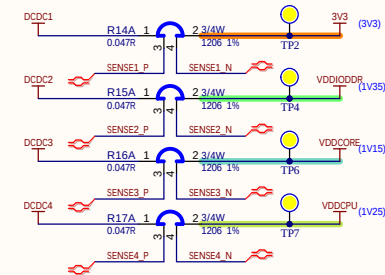
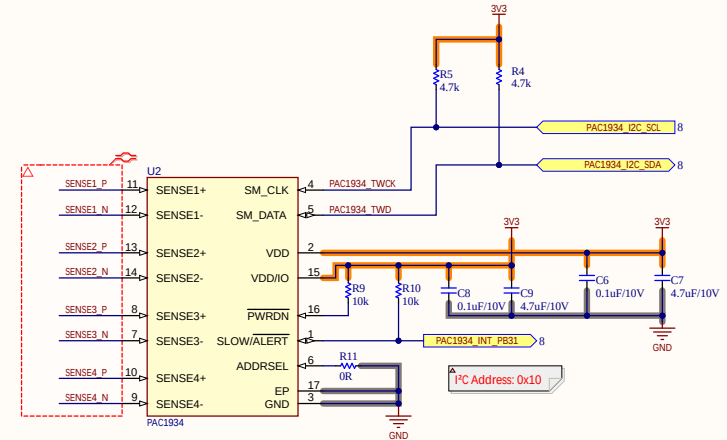
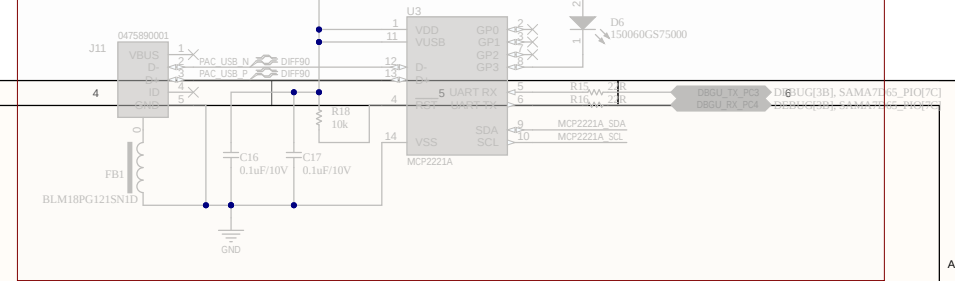
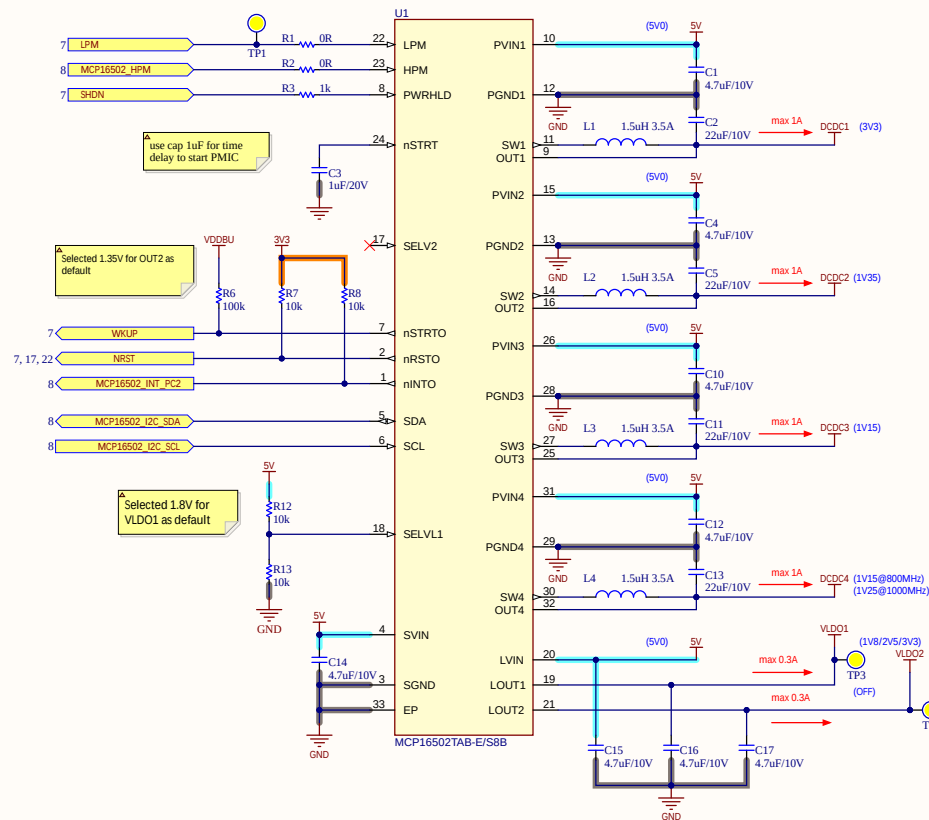
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Revision: Design: * Date: **2/7/2026**

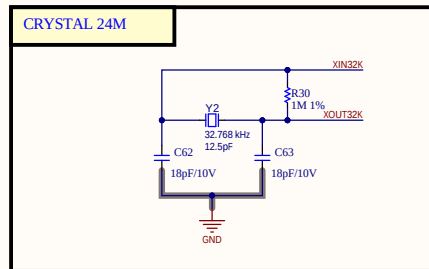
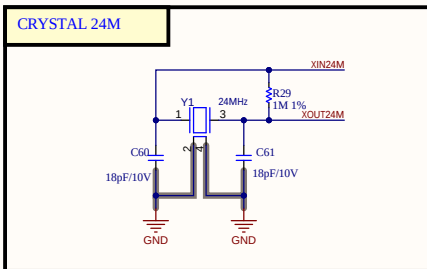
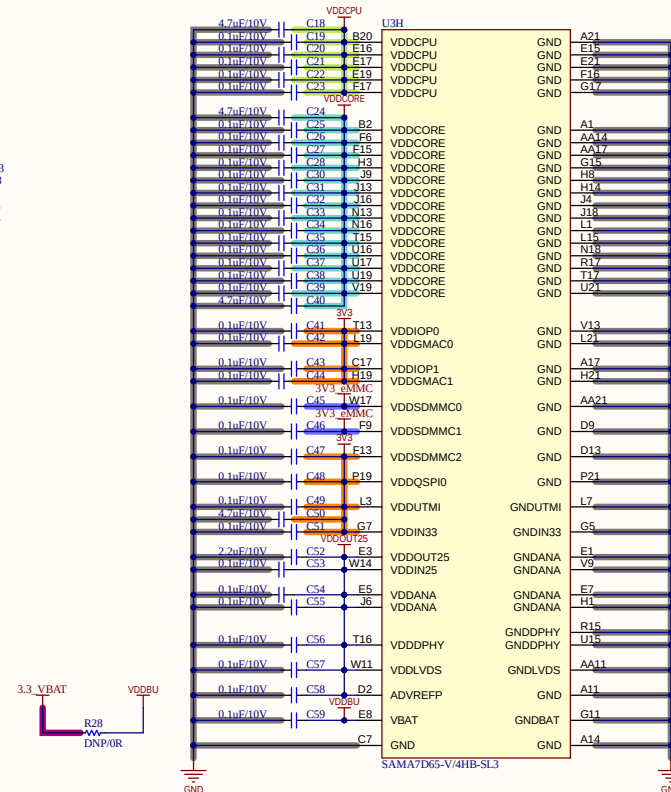
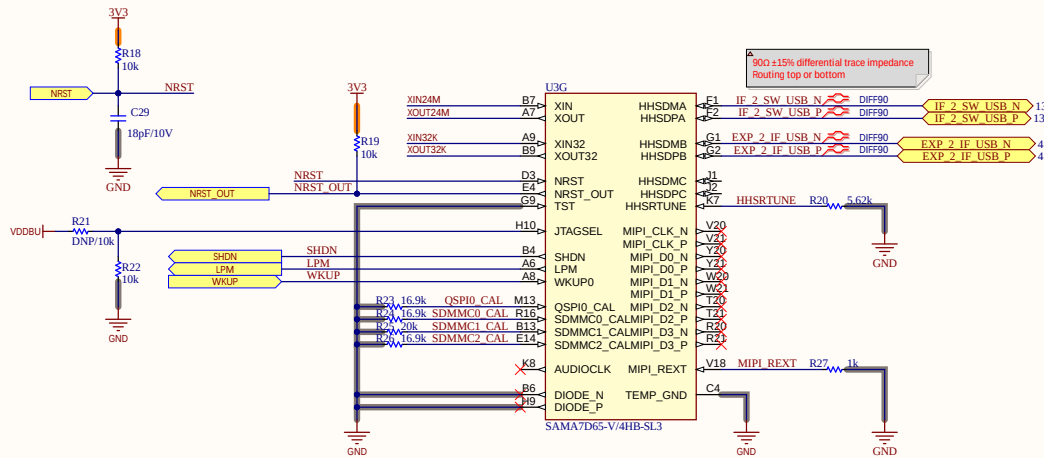
Size: **A3** Document Number: Sheet: **5** of **23**



Power Management Unit



Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER
Design by: SANG HUYNH	Sheet title: SAMA7D65_PMIC.SchDocSheet 6 of 23
Checked by: BAO BUI	Document No: 6
Approved by: VU PHAM	Revision: *
	Size: A3
	Date: 2/7/2026

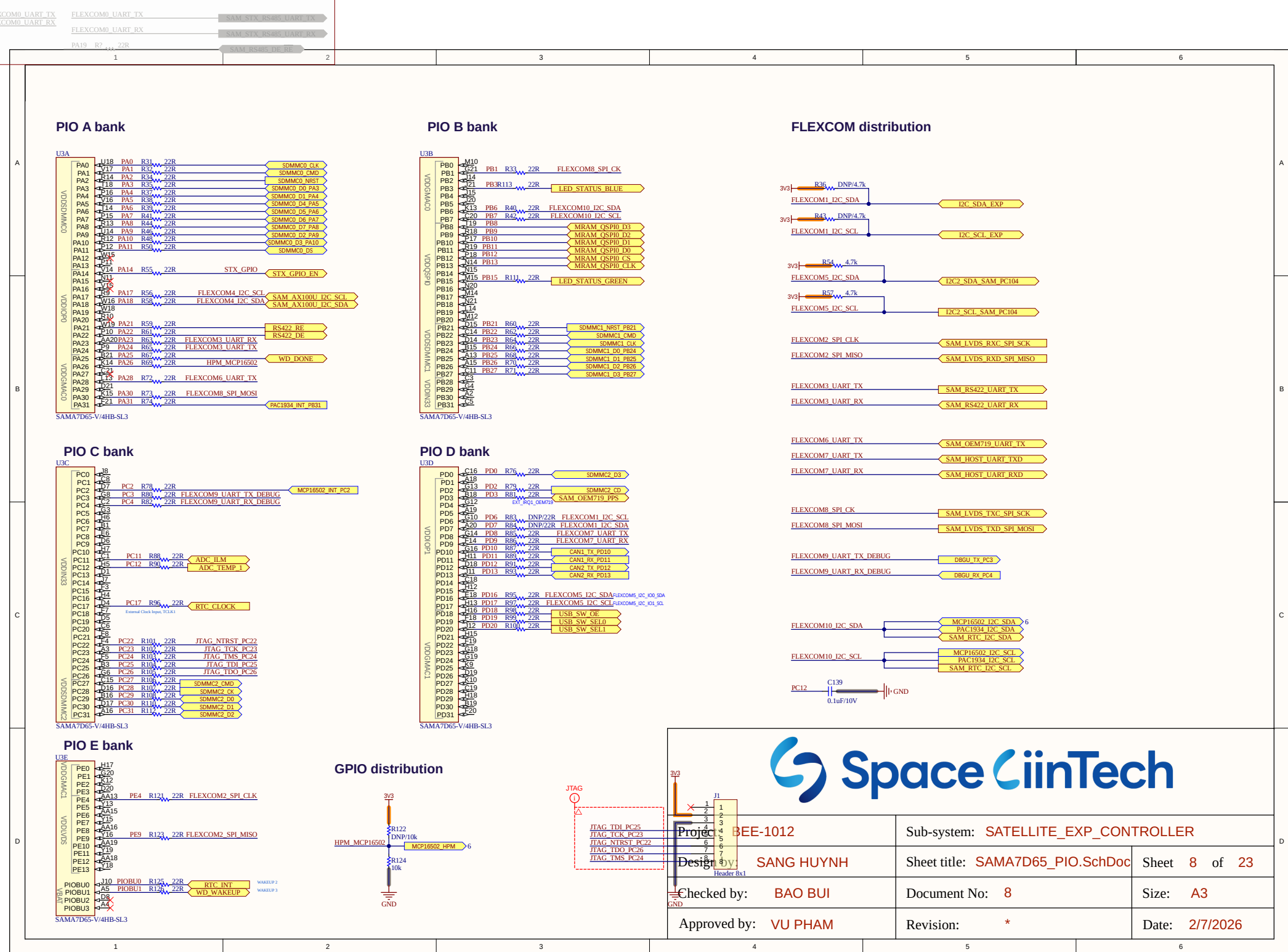


Sub-system: SATELLITE_EXP_CONTROLLER

Sheet title: SAMA7D65_MPU.SchDoc Sheet 7 of 23

Size: A3

Date: 2/3/2026



SAMA7D65 DDR3L

ADDR-CTL Matched Net Lengths [Tolerance = 10mil]

LANE0 Matched Net Lengths [Tolerance = 10mil]

LANE1 Matched Net Lengths [Tolerance = 10mil]

1000 ±10% differential trace impedance Routing top or bottom

1000 ±10% differential trace impedance Routing top or bottom

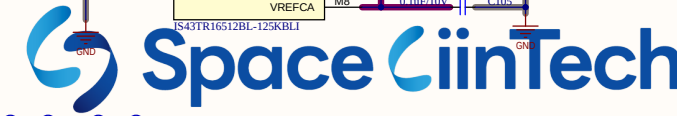
1000 ±10% differential trace impedance Routing top or bottom

1000 ±10% differential trace impedance Routing top or bottom

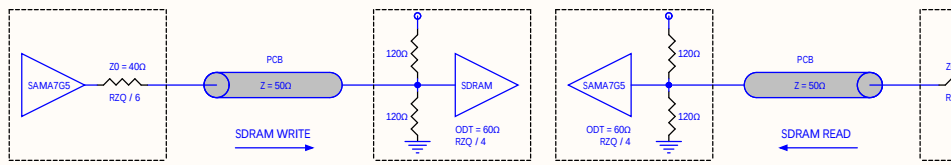
1000 ±10% differential trace impedance Routing top or bottom

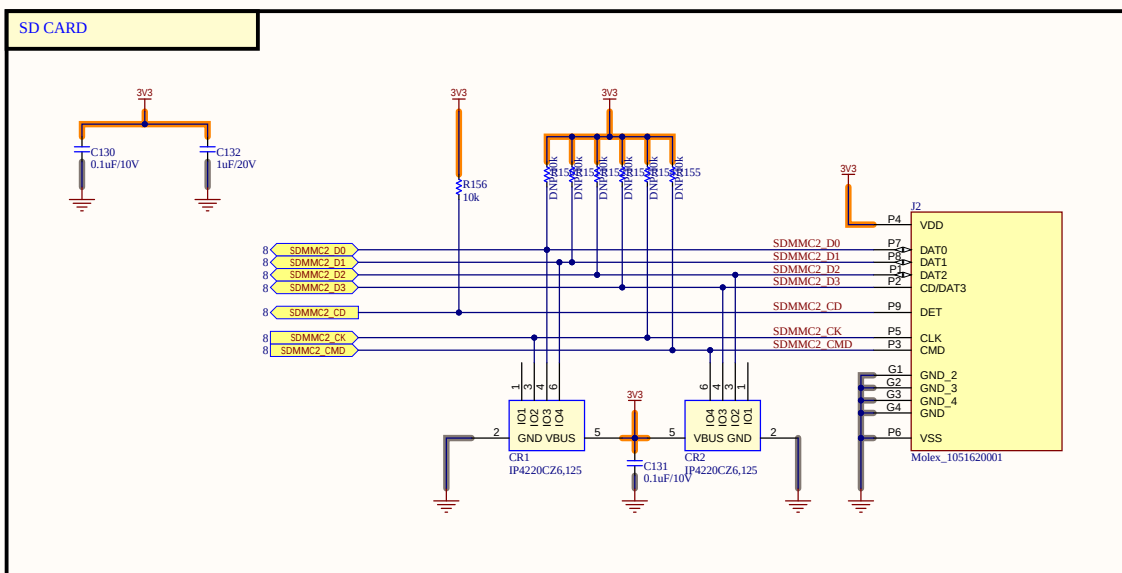
1000 ±10% differential trace impedance Routing top or bottom

1000 ±10% differential trace impedance Routing top or bottom



Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER
Design by: SANG HUYNH	Sheet title: SAMA7D65_DDR3L.SchDoc
Checked by: BAO BUI	Document No: 9
Approved by: VU PHAM	Revision: *
	Size: A3
	Date: 2/2/2026





Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: SDCARD.SchDoc	Sheet 10 of 23
Checked by: BAO BUI	Document No: 10	Size: A3
Approved by: VU PHAM	Revision: *	Date: 2/3/2026

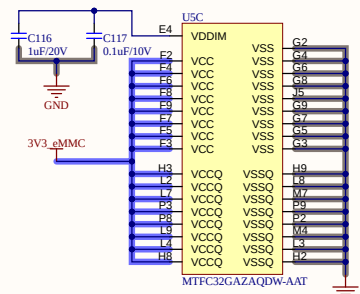
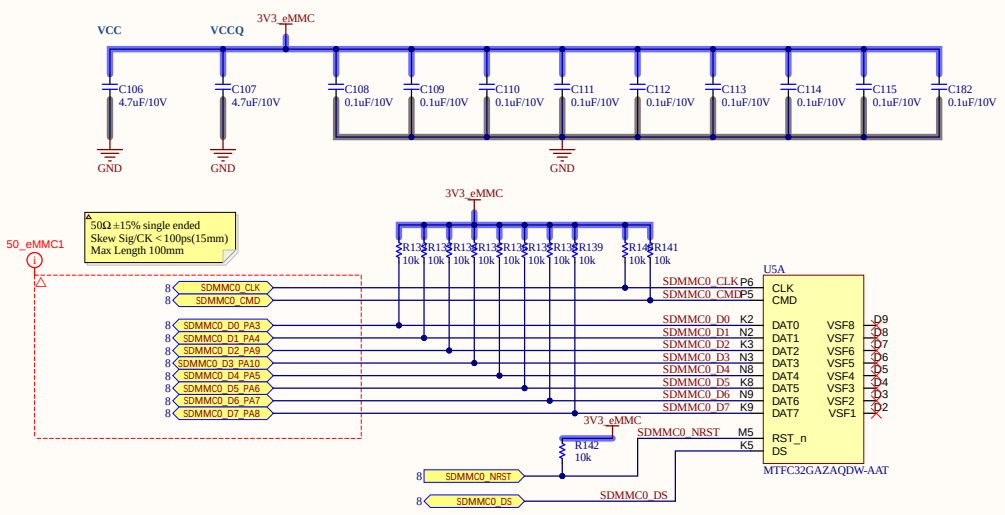
	C3	2.2	4.7	2.2	μF	7
	C4	0.1	0.22	0.1		
	C5	1	4.7	1		
	C6	0.1	0.1	0.1		

V _{CC} = 3.3V and V _{CCQ} = 3.3V nominal	V _{CCQ} supply	V _{CCQ}	-0.2	2.4	V
	Voltage input	V _{IN}	-0.6	4.6	V
	V _{CC} supply	V _{CC}	-0.6	4.6	V
	V _{CCQ} supply	V _{CCQ}	-0.6	4.6	V

Note: 1. Voltage on any pin relative to V_{SS}.

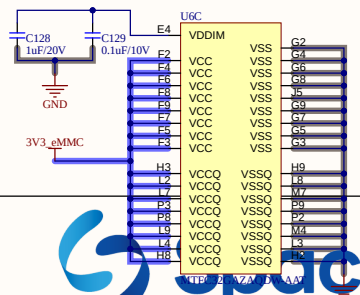
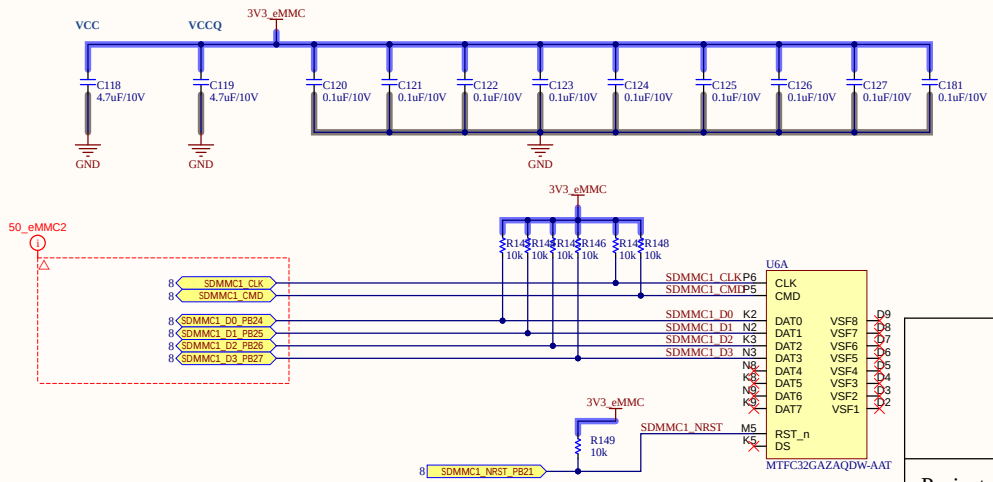
Figure

eMMC 1



USB		U10	
A1	NC	NC	U9
A2	NC	NC	U9
A9	NC	NC	U1
A10	NC	NC	U1
B1	NC	NC	U10
B10	NC	NC	U1
J6	RFU	RFU	U7
J10	RFU	RFU	U8
J1	RFU	RFU	U9
J2	RFU	RFU	U4
E2	RFU	RFU	U5
E3	RFU	RFU	U6
E4	RFU	RFU	U2
E5	RFU	RFU	U3
E6	RFU	RFU	U6
E7	RFU	RFU	U2
E8	RFU	RFU	U3
H3	RFU	RFU	U8
H4	RFU	RFU	U9
H5	RFU	RFU	U4
H6	RFU	RFU	U5
H7	RFU	RFU	U6
P2	RFU	RFU	U7
P4	RFU	RFU	U7

eMMC 2



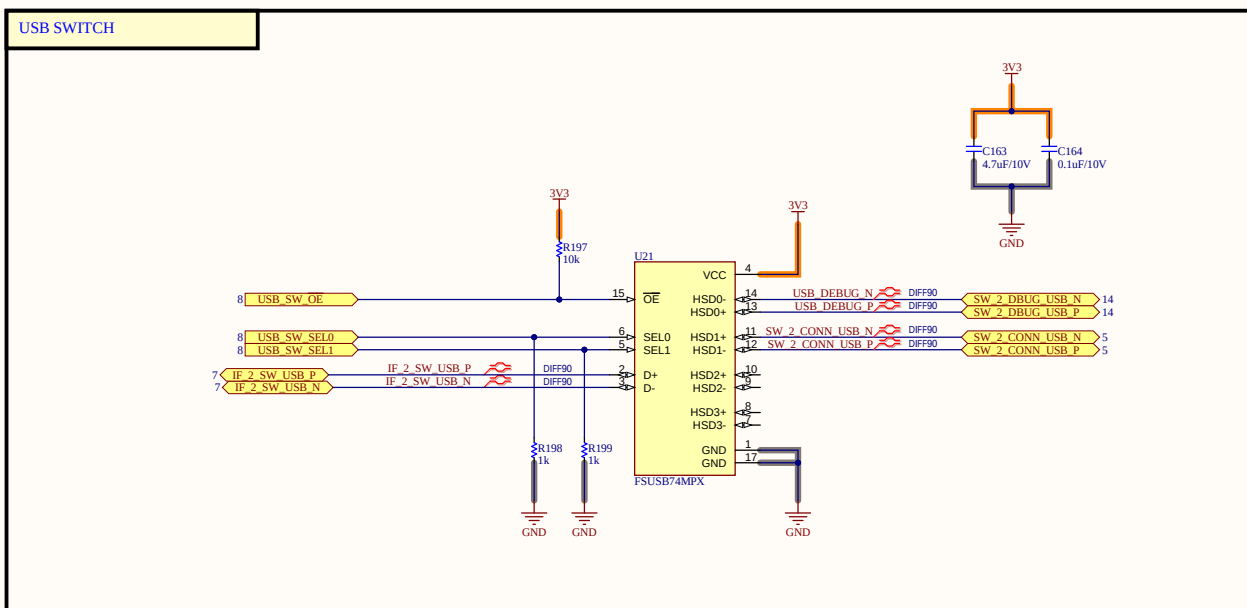
U6B		U10	
A1	NC	NC	U9
A2	NC	NC	U9
A9	NC	NC	U1
A10	NC	NC	U1
B1	NC	NC	U10
B10	NC	NC	U1
J6	RFU	RFU	U7
J10	RFU	RFU	U8
J1	RFU	RFU	U9
J2	RFU	RFU	U4
E2	RFU	RFU	U5
E3	RFU	RFU	U6
E4	RFU	RFU	U2
E5	RFU	RFU	U3
E6	RFU	RFU	U6
E7	RFU	RFU	U2
E8	RFU	RFU	U3
H3	RFU	RFU	U8
H4	RFU	RFU	U9
H5	RFU	RFU	U4
H6	RFU	RFU	U5
H7	RFU	RFU	U6
P2	RFU	RFU	U7
P4	RFU	RFU	U7

Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: eMMC.SchDoc	Sheet 11 of
Checked by: BAO BUI	Document No: 11	Size: A3
Approved by: VU PHAM	Revision: *	Date: 1/30/20

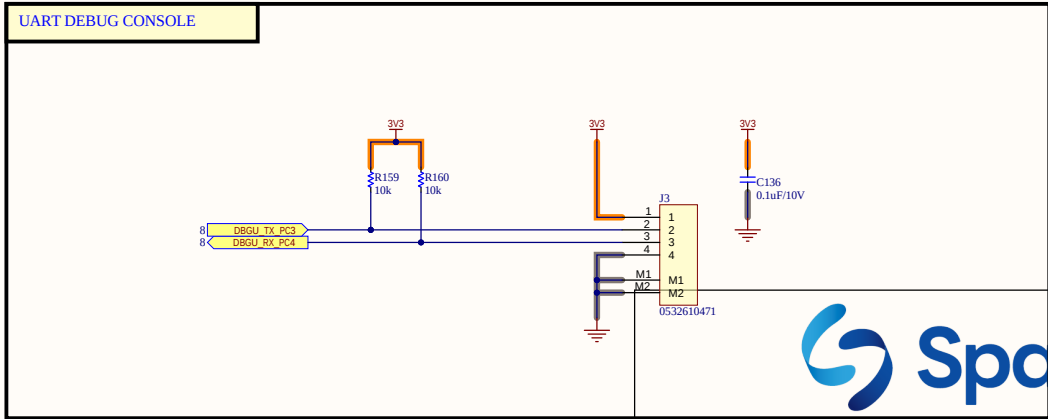
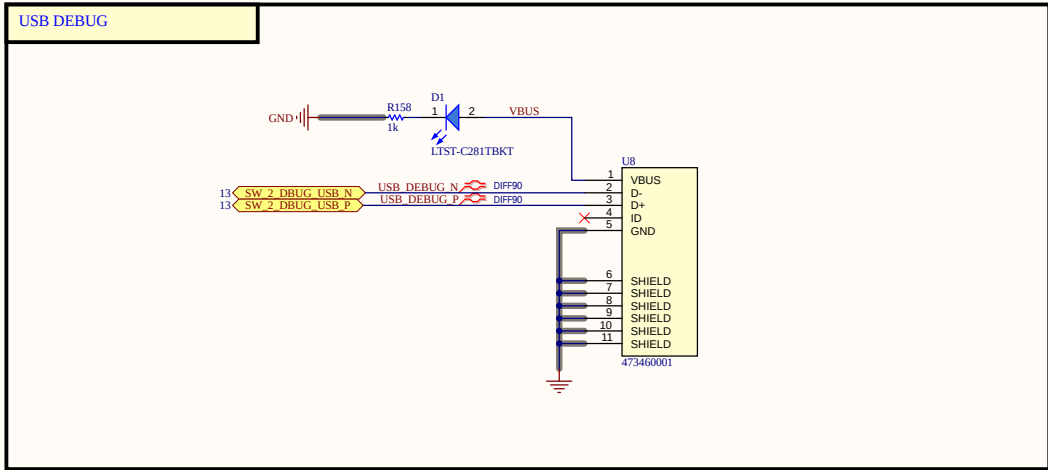
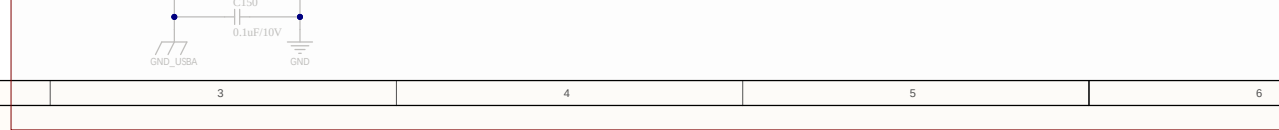
Table

Pull-up
Pull-up
Pull-up
CLK/CMD
Serial r
Serial r
Pull-do
V _{CCQ} ca
V _{CC} cap
V _{DDIM} 6

Notes:

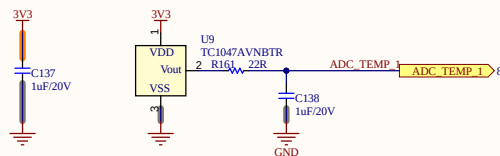


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: USB_SWITCH.SchDoc	Sheet 13 of 23
Checked by: BAO BUI	Document No: 13	Size: A3
Approved by: VU PHAM	Revision: *	Date: 2/3/2026

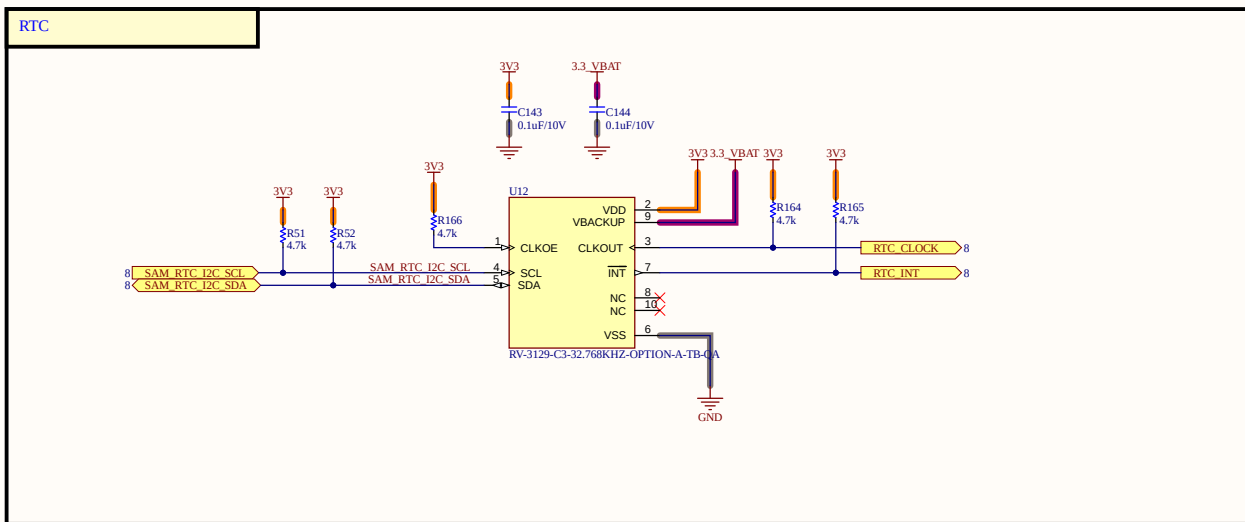


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER
Design by: SANG HUYNH	Sheet title: DEBUG_CONSOLE.SchDoc Sheet 14 of 23
Checked by: BAO BUI	Document No: 14 Size: A3
Approved by: VU PHAM	Revision: * Date: 2/3/2026

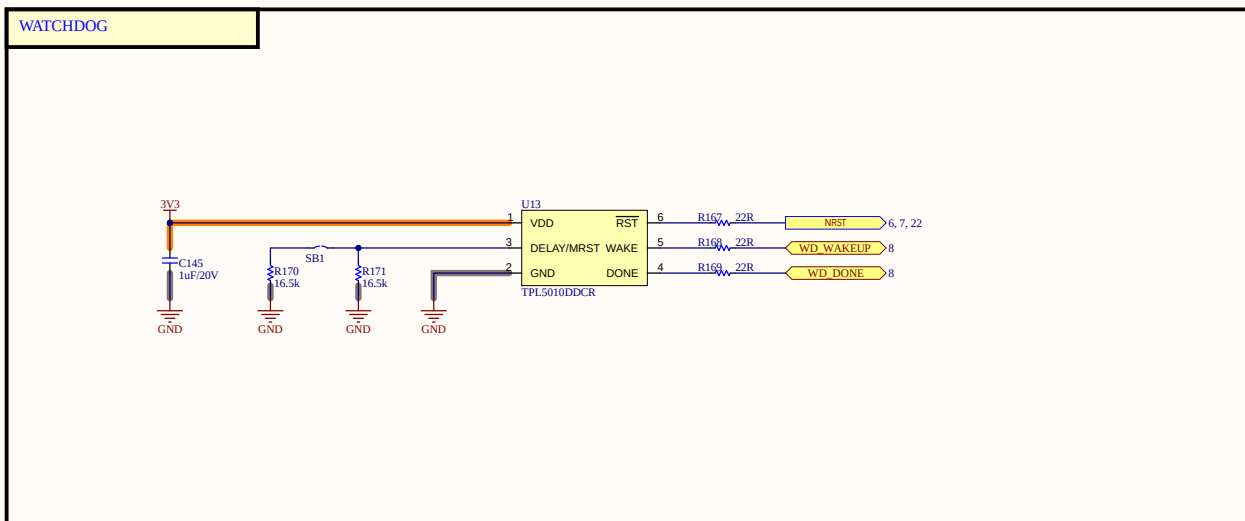
SENSOR TEMP



Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: TEMP.SchDoc	Sheet 15 of 23
Checked by: BAO BUI	Document No: 15	Size: A3
Approved by: VU PHAM	Revision: *	Date: 2/6/2026

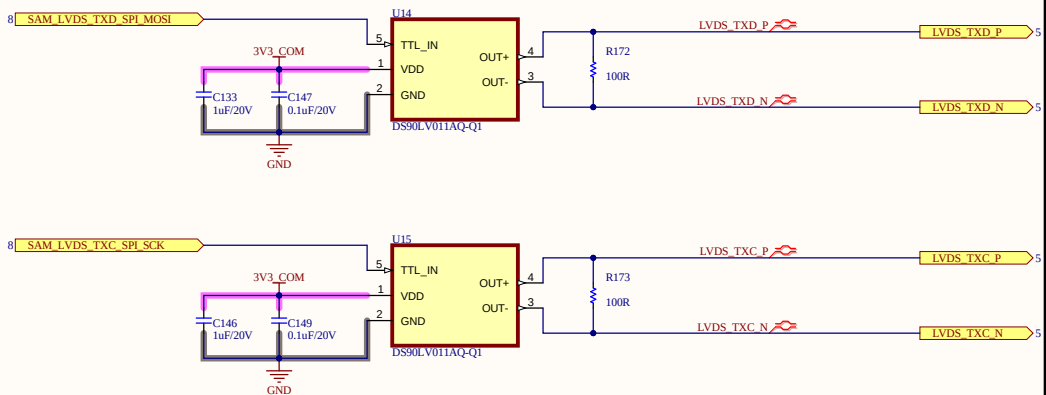


Project: BEE-1012		Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: RTC.SchDoc	Sheet 16 of 23	
Checked by: BAO BUI	Document No: 16	Size: A3	
Approved by: VU PHAM	Revision: *	Date: 1/30/2026	

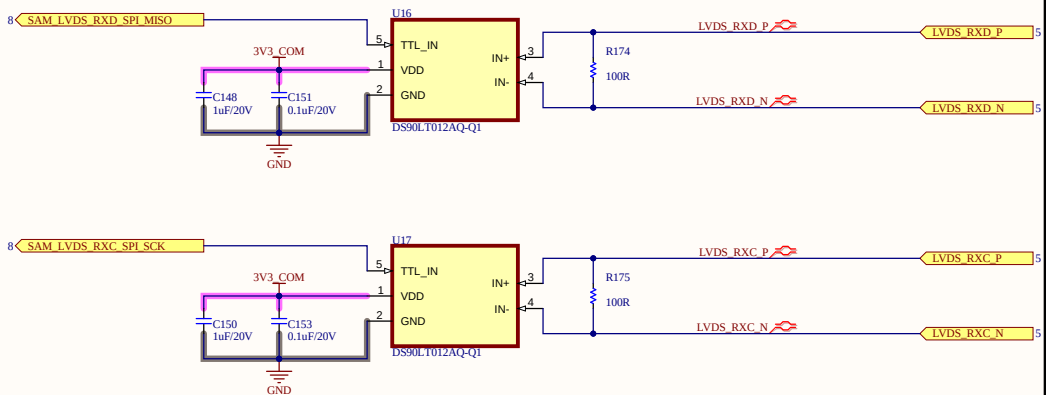


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: WATCHDOG.SchDoc	Sheet 17 of 23
Checked by: BAO BUI	Document No: 17	Size: A3
Approved by: VU PHAM	Revision: *	Date: 1/27/2026

SPI LVDS TX



SPI LVDS RX



Project: BEE-1012

Sub-system: SATELLITE_EXP_CONTROLLER

Design by: SANG HUYNH

Sheet title: LVDS.SchDoc

Sheet 18 of 23

Checked by: BAO BUI

Document No: 18

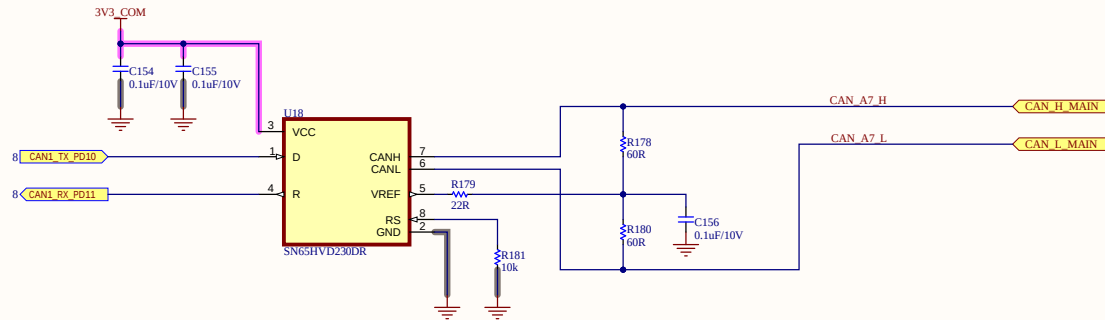
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Approved by: VU PHAM

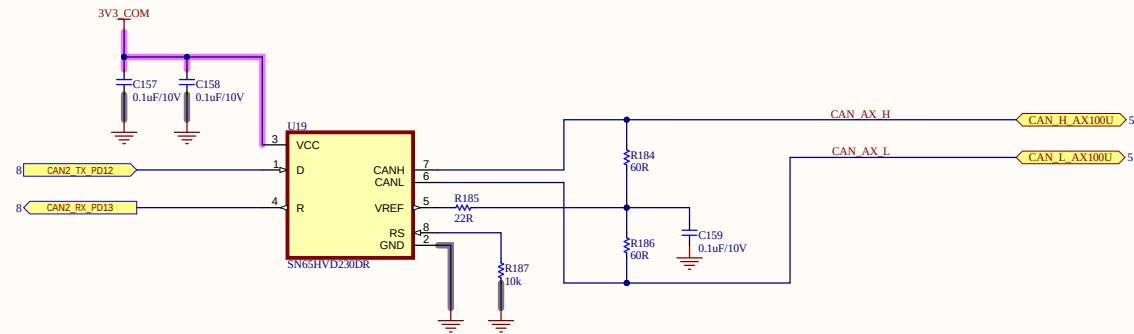
Revision: *

Date: 1/30/2026

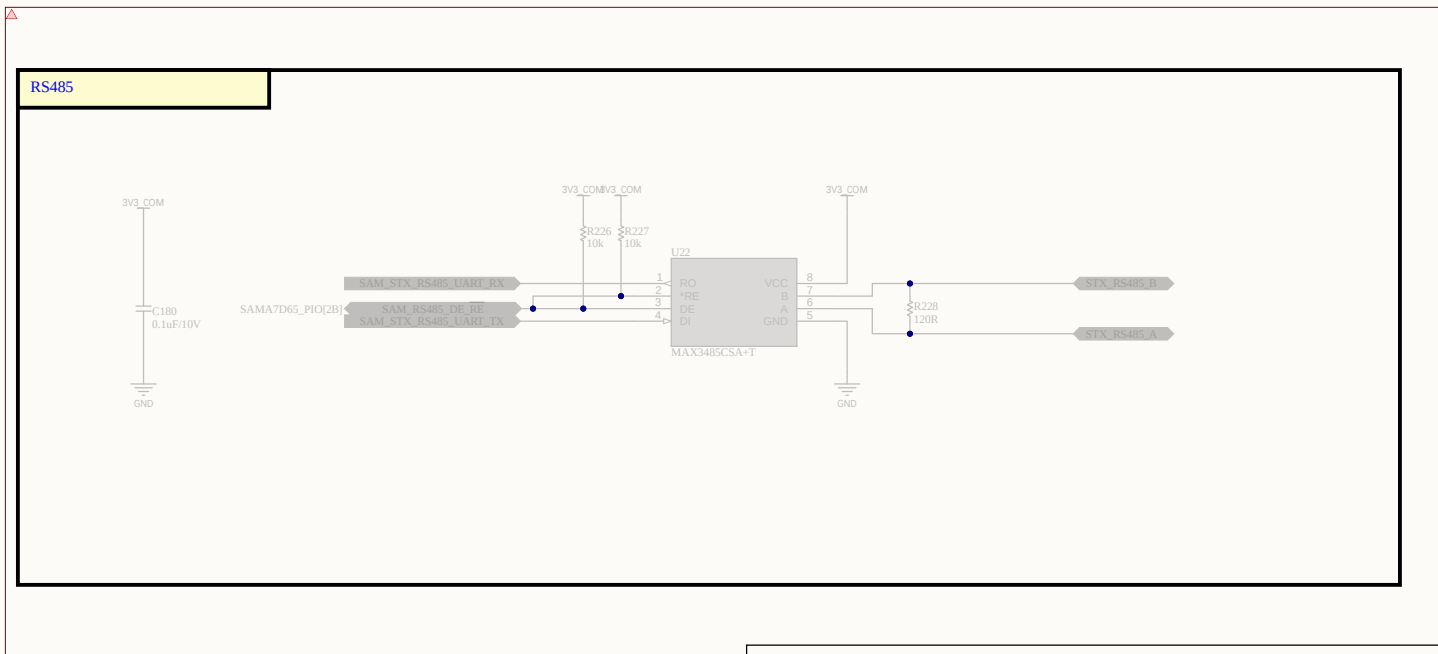
CAN1 FOR BUS CAN MAIN



CAN 2

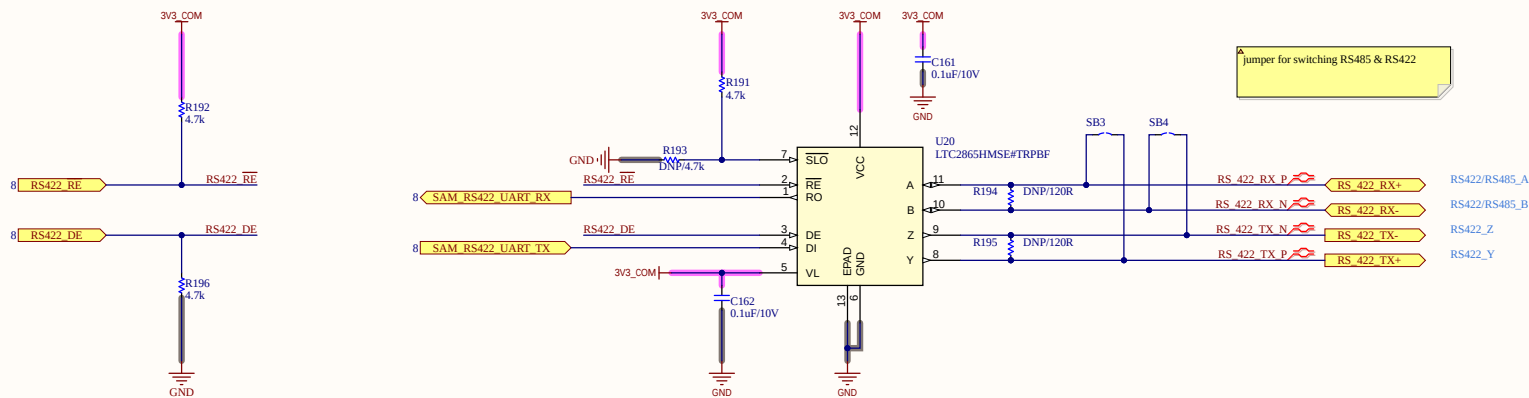


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: CAN_PHY.SchDoc	Sheet 19 of 23
Checked by: BAO BUI	Document No: 19	Size: A3
Approved by: VU PHAM	Revision: *	Date: 1/30/2026

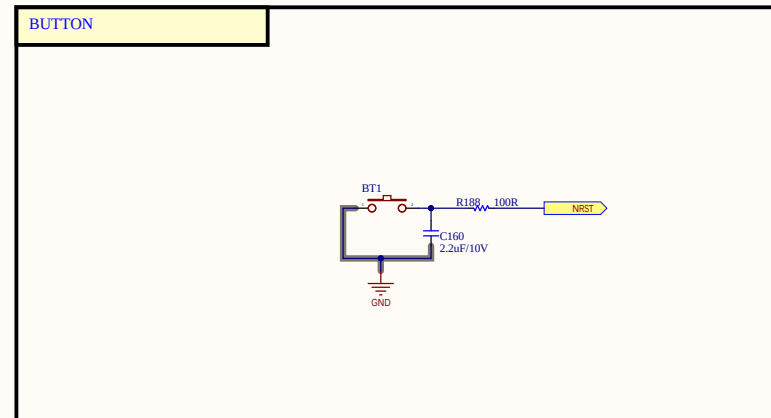
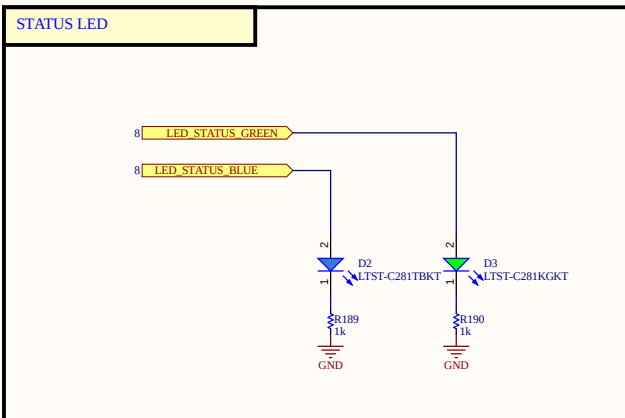


Project: BEE-1012		Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: RS485.SchDoc	Sheet 20 of 23	
Checked by: BAO BUI	Document No: 20	Size: A3	
Approved by: VU PHAM	Revision: *	Date: 1/27/2026	

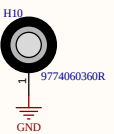
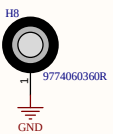
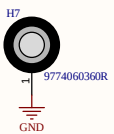
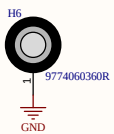
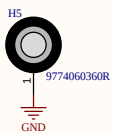
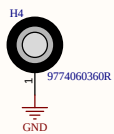
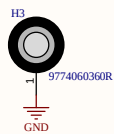
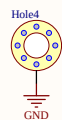
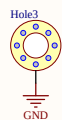
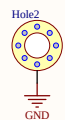
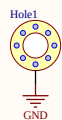
RS422



Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: RS422.SchDoc	Sheet 21 of 23
Checked by: BAO BUI	Document No: 21	Size: A3
Approved by: VU PHAM	Revision: *	Date: 2/2/2026



Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: BUTTON_LED.SchDoc	Sheet 22 of 23
Checked by: BAO BUI	Document No: 22	Size: A3
Approved by: VU PHAM	Revision: *	Date: 1/27/2026



Project: BEE-1012

Sub-system: SATELLITE_EXP_CONTROLLER

Design by: SANG HUYNH

Sheet title: MOUNTING_HOLE.SchDocSheet 23 of 23

Checked by: BAO BUI

Document No: 23

Size: A3

Approved by: VU PHAM

Revision: *

Date: 2/3/2026